Lesson Notes

MODULE 4 | LESSON 1

BEHAVIORAL FINANCE

Reading Time	60 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

So far, we have viewed investors and economic agents in general as rational economic beings. Rational economic beings are assumed to have perfect information and are capable of acting in a perfectly rational manner all of the time. But is this really the case? Over the years, psychologists have questioned this assumption, and new paradigms have begun to emerge. This has led to a shift from traditional finance to behavioral finance, which has implications for how people feel, think, and act when confronted with economic decisions. This will be the subject of Module 4, where we consider the needs and implications of shifting from the world of traditional finance to that of behavioral finance.

1. From Traditional Finance to Behavioral Finance

What we have done so far in Modules 1, 2, and 3 can be referred to as **traditional finance** (**TF**). What do we mean by that? It is very simple: in TF it is assumed that people are rational and risk-averse and that they act to maximize their own utility. In particular, the latter point means that individuals do not consider items like social values, unless these values can contribute to increase their own personal utility, that is. The typical individual in TF is considered a rational agent, called a **rational economic man (REM)**. The traits that characterize an REM are:

1. possessing perfect information, and
2. acting in a perfectly rational manner .

What does it mean to act in a perfectly rational manner? It means that individuals with a limited budget will select a mixture of goods and services to maximize their utility function. A utility function is a system of preferences that conform to the following 4 axioms:

- Given the choice among two portfolios or goods, A and B, an individual is always in a position to say whether they prefer A or B or is indifferent between the two (**completeness**);

- If portfolio A is preferred to B and B is preferred to C, then A is also preferred to C (**transitivity**);
- If portfolio A and B are mutually exclusive and A is preferred to B, then for any additional portfolio C and a number $\gamma \in (0,1], \gamma A + (1-\gamma)C$ is preferred to $\gamma B + (1-\gamma)C$;
- If A is preferred to B and B is preferred to C, there is a combination of A and C for which the individual is indifferent to B (**continuity**).

Note: The independence axiom *may* be dependent on the setting. An individual may like ice cream more than ketchup. But it is unlikely that they would like ice cream + fries more than ketchup + *fries*! And yet, this may not always be true.

The behavior of an REM leads to efficient markets, i.e., markets in which prices reflect all the relevant information available.

What happens when an REM gets a hold of new information? An REM has beliefs, and when presented with new information, they adapt their beliefs about the probabilities of uncertain events via Bayes' formula (in this sense, recall the Black-Litterman model as a sophisticated example). Hence, a REM must know the Bayes formula and be capable of performing the required calculations.

As it was already stated above, TF is also characterized by individuals who are risk averse. We know from Module 1, Lesson 3 that this means that their utility function is concave.

Behavioral Finance (**BF**), unlike TF, **does not assume** that:

- individuals are always risk-averse;
- individuals use Bayes' formula;
- individuals have perfect information;
- individuals act to maximize only their personal utility.

This does not mean that individuals cannot be REM, but they need not be all the time. Instead, the key assumptions of BF:

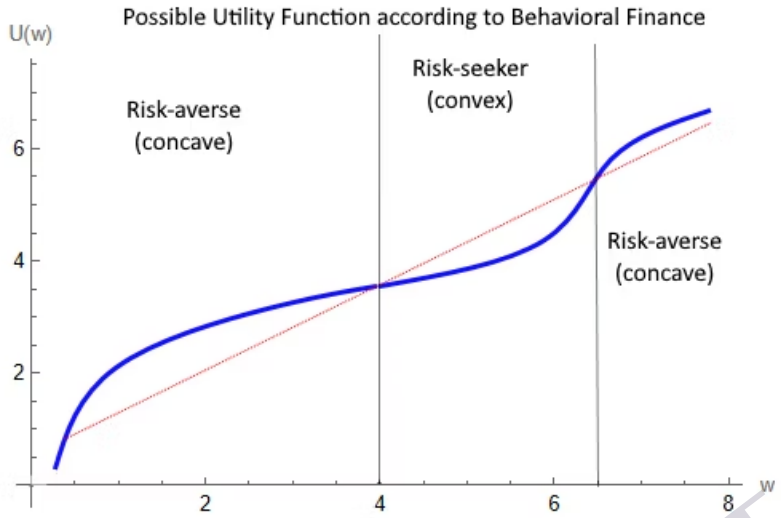
- individuals may not have access to full information;
- individuals may lack the mathematical ability to incorporate full or partial information in their decision making process;
- individuals may not be able to attain the highest utility because they are more interested in short-term goals as opposed to overall goals.
- individuals may consider philanthropy or just simple acts of kindness as useful while these have no place/value in the world of TF, which is ruled by self-interest.
- individuals' utility functions that are not always concave. Utility functions can be concave, convex (risk-seeker) or a mix of concave and convex.

In essence, the behavior of individuals in BF cannot be reduced to that of perfect calculators and their decision making is better explained by psychology.

2. From Utility Maximization to "Satisfice"

TF is based on utility theory and in particular concave utility functions (i.e., diminishing marginal utility). BF allows for more complex types of utility functions since it allows individuals to be both risk averse and risk seekers. In BF, an individual may exhibit a utility function like in Figure 1:

Figure 1: The Double Inflection Utility Function (Friedman and Savage)



In the first region, the individual is risk averse, i.e., they buy insurance in the language of utility theory we developed in Module 1. In the second region, the individual buys lottery tickets. In the third region, they are back to buy insurance. An REM would not behave like this. However, an individual driven by perceptions may decide that with wealth between 0.5 to 4 (in the picture of figure 1), they may need to protect it. When wealth is between 4 and 6.25, they may feel that there is enough wealth to take care of a certain amount of basic needs, and it is fine to gamble some to achieve a higher level of wealth still. Once wealth is above 6.25, the individual is back to being risk-averse because perhaps they have reached a very desirable level of wealth to fulfill their basic needs and some additional aspirational goals as well and feels that now wealth must be protected lest they lose the ability to fulfill such aspirational goals. It should be evident that we are assuming that individuals are driven by their own psychological perception of wealth in terms of goals that they can or cannot realize.

Is BF supported by evidence? The answer is yes. For instance when individuals suffer from anxiety, irritability, or depression they exhibit a certain type of behavior towards losses or risks. On the other hand when they feel euphoric they display a different attitude toward losses and risk. Some studies have pointed out that this may be a matter of our bodies producing serotonin or dopamine although the link is not accepted by everyone. It has also been pointed out that when an individuals anticipate rewards, they produce more dopamine and if they do not achieve certain results, then the dopamine level goes down.

The science that tries to establish a relationship between the behavior of investors and the way the brain works is called **neuronomics.**

TF has no explanation for asset bubbles, for instance. However BF can explain it. TF has no explanation for the equity risk premium puzzle: Rajnish Mehra and Edward Prescott in their 1985 paper found that the average real annual yield on equity in competitive pure exchange economies should be “a maximum of 0.4% higher than that on short-term debt.” However, they found that historically it stood at 6%. BF can provide some justifications for this, and there are more instances that highlight the fact that BF is a real thing and not just a nice theory.

Having said what we just did, it is useful to explain that the attitude towards BF is not uniform among practitioners and academics. To this purpose, Nassim N. Taleb wrote in *Statistical Consequences of Fat Tails*, on page 12, that:

"*Examples of fields that are, practically, frauds in the sense that their results are not portable to reality and only serve to feed additional papers that, in turn, will produce more papers: Modern Financial Theory, econometrics (particularly for macro variables), GARCH processes, psychometrics, stochastic control models in finance, behavioral economics and finance, decision making under uncertainty, macroeconomics, and a bit more*."

Clearly, Taleb kills many branches besides behavioral economics and finance in his statement. Nevertheless, several prescriptive approaches (i.e., approaches that are not derived by theories but by observing how investors actually make decisions) have been studied. We could call these approaches **heuristic approaches**, which were pioneered in the 1970s and 1980s by Kahneman and Tversky (see the References for their seminal 1979 paper in *Econometrica*). Daniel Kahneman, a psychologist, was awarded the Nobel Prize in Economic Sciences in 2002. Amos Tversky, a mathematical psychologist, died in 1996 but Kahneman acknowledged in an interview with the Stanford Report (find the interview here) that "*The prize ... is quite explicitly for joint work, but unfortunately there is no posthumous prize*." In 2017, Richard Thaler was also awarded the Nobel Prize in Economic Sciences "for work that had explored how human psychology shaped economic decisions."

In the beginning, many economists rejected behavioral finance as experiments lacking clear links to real life; however, later, the field has gained more acceptance. In a 2010 commentary (link to commentary), Kenneth Arrow, co-winner of the 1972 Nobel Prize for Economic Sciences wrote that:

"*We see, by the way, that today psychology is invading economics—the whole field of behavioral economics. I believe that sociology should play more of a role in economics than it does. The way people behave in economics is partly influenced by how other people behave. It's easy to point out examples, but it's not so easy to construct a broad theory*."

If REM is the typical individual for TF, the prototype individual of BF exhibits **bounded rationality**. Essentially, individuals have limits in their knowledge capacity and the assumption of perfect information must be removed. Individuals, instead of maximizing their utility, content themselves with satisfice, that is individuals operate with partial information and use heuristics instead of mathematical theories to process their information and transform them into decisions.

Example (satisfice). Say that you were hired for a job and need to relocate to a new city. Among other things, you will need to find an apartment or a house to live in, get utilities connected, find a bank, etc. Let's concentrate on finding a bank. As an REM, you would go all over town, find out everything you need to know about every single bank, and then decide which bank is the best one for you. Proximity may be taken into account in your utility function as much as interest rates, fees, and types of transactions the bank can execute for you. However, collecting all the information is quite a chore. The BF individual will probably visit a few banks and then decide based on that partial information that one of those rates offered is adequate and probably would not attach a lot of utility to obtaining more interest rates and going all over town for several more days. This is an example of bounded rationality and satisfice. Same is true for the housing unit. Will you visit a certain number of units before deciding on one, or will you visit every single available unit before making your decision?

Example (satisfice). Max Schmelling has a new client, Jon, who is interested in becoming an antique car investor and is ready to spend $200,000 from his portfolio to buy the first car for his collection. Shortly afterwards, the client visits an antique car auction not far from his home Unfortunately, the auction had a limited number of cars meeting his requirements. He is subsequently drawn to one antique car in particular, even though it does not have several of the features he wanted. Nevertheless, after some consideration, he decides to purchase it anyway. and the car is now in storage for safekeeping. This is an example of *satisfice*.

Exhibiting bounded rationality should not be considered a pejorative form of rationality. Bounded rationality takes into account that it is often not practical to consider all possible outcomes or that computations may become too complex. In the contest of portfolio theory, while an REM will strive for the optimal solution, the "satisfice man" will be fine with a satisfactory solution, albeit suboptimal.

3. Prospect Theory

As already stated, bounded rationality removes the assumptions of perfect information and maximization of expected utility. Prospect theory goes one step further and replaces the assumption that individuals are risk averse with another assumption according to which individuals **exhibit loss aversion** instead. In particular, individuals fear losses more than they enjoy gains. In other words, **investors focus on change in wealth** rather than on wealth per se, and given a potential loss and gain of equal amounts, the increase in utility associated with the gain is smaller than the disutility associated with the potential loss. Note the subtle points: It is not just that economic agents have positive or negative change in values of utility from gain or losses; respectively, the change in the case of losses are larger than the changes associated with gains.

Experiments have shown that most individuals if offered a gamble with 50/50 odds of equal but opposite payoffs, they will not take the gamble. The only explanation for this is that they overreact to small probabilities and underreact to large probabilities, i.e., probabilities have a psychological component as well. A 1% probability of losing, for example, might be perceived as 10%, and a 99% probability of winning is felt as if it were, say, 90% instead.

Finally, individuals experience diminishing sensitivity applied to upward and downward perceptions and to evaluation of changes of wealth. This means that, given a set of possible outcomes $\{x_1, \ldots, x_n\}$ with probabilities $p_1, \ldots, p_n$, the utility function in prospect theory looks like:

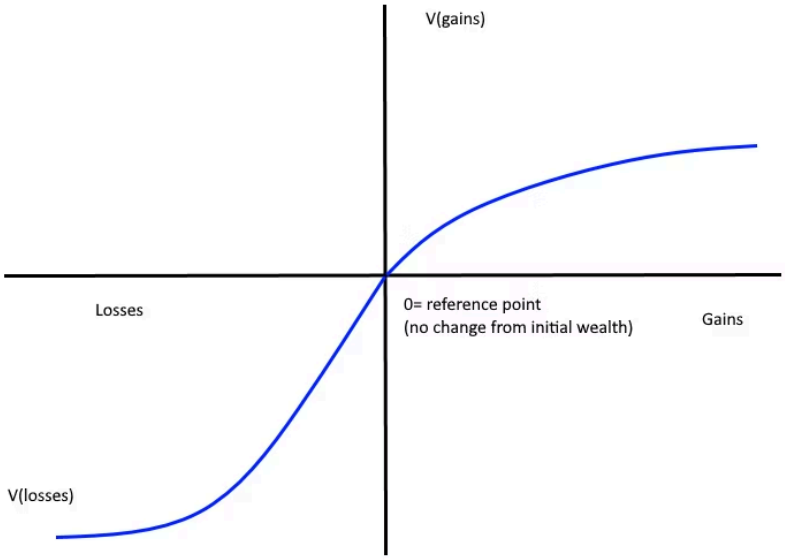
$$U(x_1, \ldots, x_n) = w(p_1) \cdot p_1 \cdot V(x_1) + \ldots + w(p_n) \cdot p_n \cdot V(x_n)$$

where $w(\cdot)$ is a weighting function assigned to probabilities. In TF, it would be $w(p) = p$, while in prospect theory, it can be something else.

Also, we assumed n possible outcomes, but the number of outcomes can be infinite. $V(\cdot)$ is a value function that is based on changes and not on wealth so that $V(x_i)$ represents the change in wealth if outcome x_i is the result.

To reflect the descriptions of prospect theory, the function V must look something like in figure 2, i.e., S-shaped and asymmetrical.

Figure 2: The Function $V(\cdot)$ is S-shaped and Asymmetric



Example. Consider the following situation in the context of prospect theory:

Outcome	value	probability	weight(probability)
x1	-100	3%	1.50
x2	-35	5%	1.20
x3	-1	25%	0.90
x4	30	30%	0.80
x5	90	37%	0.75

Then, the utility function can be calculated as

$$U = -100 \cdot 1.50 \cdot 3\% - 35 \cdot 1.2 \cdot 5\% - 1 \cdot 0.90 \cdot 25\% + 30 \cdot 0.80 \cdot 30\% + 90 \cdot 0.75 \cdot 37\% = 25.35.$$

This psychological approach warns us about the fact that investors fear losses and can become risk seekers in an attempt to avoid losses despite being risk averse when presented with gains. Empirical studies have shown that:

- on average, if individuals are presented with a gamble where they have an equal chance to win $100 or lose $70, they will not take the gamble;
- on average, if offered a sure loss of $75 or a gamble where they have 50/50 chance to win $30 or lose $200, they take the gamble even if its expected value is -$85.

Prospect theory can explain why investors tend to concentrate on high-risk and low-risk investments and do not seem to have much of an appetite for medium-risk assets.

To understand why individuals behave the way they do, one has to dig into the psychological causes of these behaviors, which will be the subject of the next section.

In the meantime, Table 1 summarizes the key differences between TF and BF.

Table 1: Summary of the Differences between TF vs. Bounded Rationality and Prospect Theory

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Traditional Finance	Bounded Rationality+Prospect Theory
Perfect knowledge	Limited knowledge
Utility is maximized	Satisfice
Decision making is fully rational	Limitations on cognitive ability when making decisions
Risk aversion	Loss aversion

4. Behavioral Biases of Individuals

Individuals are called upon to make thousands of decisions every day. Traditional finance's REM follows the fully rational steps. However, many individuals, when facing complex decision-making situations, often either lack the time, the ability, or both to follow an optimal course of action and are prone to take shortcuts and/or follow their intuition or choose to utilize simple rules of thumb. Furthermore, most individuals are also prone to let emotions and social influences affect their decision-making process.

An error is something that happens at random; a bias is something that is systematic and thus predictable. We cannot do much about unpredictable errors; however, BF tells us that individuals suffer from many biases. For convenience, they can be grouped in two main groups: **cognitive and emotional biases**.

- cognitive biases are the result of faulty reasoning and/or the inability of individuals to follow and/or understand complicated arguments (statistical techniques, processing information, memory errors, honest mistakes, etc.);
- emotional biases are caused by individuals' feelings and impulses or by following one's intuition and, as such, are not completely conscious.

Most individuals suffer from a combination of both groups. It is not always easy to distinguish what is what. However, we should make an attempt because a cognitive bias can be eliminated or reduced via better training or education; an emotional bias, on the other hand, is more difficult to remove and often has to be accommodated.

It would be a mistake to think of these biases as something completely theoretical. Real individuals do not behave like textbook individuals.

Example: Dealing with customers as a portfolio manager. Suppose you are assigned a customer, a 69-year-old woman who inherited a considerable amount of money from her late husband. Her late husband served as an executive for an oil company for over three decades and left her a considerable amount of wealth, mostly tied in oil stocks. This person comes to your office to discuss her portfolio. You explain to her that she is in a very good situation but that you are concerned that nearly 75% of her holdings are tied up in oil companies and that the oil industry is not expected to do particularly well over the next 2–3 years. Immediately, she shuts you down: "I would not be honoring the memory of my late husband if I were to sell the assets he left me, so these assets cannot be touched!" This is clearly a situation in which a person exhibits an emotional bias. It also appears that it would prove nearly impossible to remove it, since she feels so strongly about it. The best thing you can probably do is to try working with her instead; perhaps consider moving some assets around but within the oil industry sector. She accepts, as long as you keep her investments in oil companies. This is an example of moderating a bias. Had you insisted on

changing the composition of her portfolio dramatically or that she reduce her exposure to the oil sector, that action could have led to an unhappy customer or even to a customer firing the firm. Portfolio managers have to earn a living too.

These situations happen more often than you might think. In another instance, you may find yourself dealing with a 50-year-old man who created his own company and has done very well for himself. He has a sizeable portfolio consisting of mostly tech stocks. He has no problem changing the composition of his portfolio but has one demand: high returns and avoiding "boring investments like those his father makes." Again, this is another individual who exhibits emotional biases mostly. He was smart enough to understand your objections when you point out that his portfolio is very risky, but he feels that he knows when to get out of it if necessary. Again, you will be forced to attempt to moderate his biases by proposing tech stocks that you and or your analysts perceive as safer than several of the names he wanted to hold. Trying to correct his excesses would not sit well with a person who feels they can do anything.

Finally, a third case concerns a 60-year-old woman who is a senior executive in the fashion industry. She also has an excellent portfolio accumulated over a period of many years but is returning only 2–3% a year. You explain to her that, without increasing her risk much, she could achieve better returns. She becomes sort of frantic and explains to you that she is very worried about losing money as she plans to live off it in retirement and provide for her sick sister. However, after you showed her several simulations and she realizes that the additional risk is not substantial, she gives you the mandate to change the composition of 1/3 of her portfolio. This is a case of a person who had cognitive errors (she could not understand the simulations and what they meant) but once you explained them to her, she became open to the possibility of changing her portfolio's composition.

What is the moral of these short stories? One cannot always expect to be dealing with an REM and has to learn to deal with real people and their strengths and weaknesses, or they can lose a client. It's very possible that you, as a quantitative analyst, will end up in this position, and losing clients is not a good thing. These concepts are not just theoretical; they happen in real life.

The following tables list the most typical cognitive and emotional biases and provide general descriptions.

Table 2: Cognitive Biases

Type of bias	Description of bias
conservatism	information used to make the original decision/forecast is considered more relevant than new information
confirmation	looking for data that support one's beliefs and discounting information that contradicts one's beliefs
anchoring	reluctance to modify opinions or decisions, even when the information makes those conclusions unlikely. The individual is locked on certain ideas.
illusion of control	individual feels that they have control over outcomes when in reality they have little or none at all
hindsight	selective memory; i.e. indivdidual remembers past events where she was right but forgets instances where she was wrong

Type of bias	Description of bias
framing	information is treated differently depending on how it is presented or obtained.
representativeness	assuming that an original classification and methodology works in all cases. Sometimes it is described as deducing too much from small samples of new information.
mental accounting	considering multiple investments as separate entities and not as an overall portfolio.
availability	memorable events are given more weight in changing the probability of events than less memorable ones.

Table 3: Emotional Biases

Type of bias	Description of bias
loss aversion	assigning more (negative) value to losses than (positive) value to gains of same amount
overconfidence	illusion of having better information than others or a superior ability to "read the markets"
self-control	lack of discipline, favoring short-term goals over long-term ones
self-attribution	belief that one's successes are due to their own character and abilities, rather than luck or situational factors
status quo	preferring one's environment and/or situation to new ones, described also as reluctance to change(s).
endowment	valuing what one has more than something they do not own
regret aversion	doing nothing for fear that decisions could turn out to be wrong

Example. Tina has just received new information regarding her investment in SuperStar Company. She forecasted a return for SuperStar of $10\%.$ The new information seems to point to a potential annual return of $4\%,$ instead. However, she is unwilling to use the new information to revise her forecast. Which particular bias is Tina displaying? How would your answer change if Tina decided that, despite the new information, she still believes a10% return is achievable?

This is a clear case of conservatism bias because Tina is clearly assigning more weights to the original information used to forecast a10% return rather than the new information. In the second case, the fact that, despite the new information, she feels that her original forecast should still stand resembles more an instance of anchoring bias as she does not feel like she should modify her original forecast.

Example. Jayne set up her initial asset allocation in her defined contribution plan over 20 years ago. She was then 25 years old and fresh out of graduate school with limited financial education. She decided to put an equal amount into each asset. Over time, she has increased her contribution to the plan. Her portfolio did well and at age 47, she has a portfolio valued at $1.3 million while her asset allocation has not changed much over the period. What bias does Jayne display? Could it be corrected or should it be just moderated?

There is nothing particularly outrageous about Jayne's portfolio. The only elements we are given is that she invested the same amount in each asset over 20 years ago and the weights in her portfolio have not changed since. This makes for a textbook case of status quo bias since the investor leaves the weights unchanged and does not change them in response to changing market conditions. Since status quo bias is an emotional bias, it is difficult to correct, and it is more of a candidate for moderation.

Example. Max inherited substantial wealth from his family a number of years ago. He is now 55 and devotes his time to philanthropic activities. He comes to Jonas as a new client referred by a mutual friend. During their first meeting, Jonas has noticed that the majority of Max's wealth is concentrated in the stock of a large industrial company that his grandfather cofounded. Jonas explains his concerns about risk and advises Max to replace this stock with a more diversified and efficient portfolio. Max is very agreeable. However, when Jonas warns Max that changing the portfolio composition requires selling a large amount of assets and that this will trigger a large capital gains tax, Max becomes reluctant to sell. Which type of bias is Max most likely displaying?

This is a case of framing bias. In fact, Max's attitude towards selling the position to invest in a new portfolio changes depending on how the decision is framed. Max is fine moving to a more diversified and efficient portfolio. However, he becomes reluctant to do so when presented with the tax consequences of selling the current stock position.

Example (Lessons from student's life). Mental and cognitive biases go beyond investing and are a reality of daily life.

- Many students fail their exams because they do not exercise sufficient self-control to study enough.
- Students exhibit overconfidence when, because they have passed previous exams, they feel they can also pass later exams without too much sweating.
- Students display representativeness bias when they assume that the way they studied for their Financial Markets class will work for their Derivatives class as well.
- Students exhibit self-attribution bias when they procrastinated studying before a hard assignment (self-control bias) and have now only a few days to work on it. Students then receive low scores and blame the fact that they merely ran out of time for the poor performance instead of recognizing their inability to organize their work or their lack of skills.

5. Conclusion

This lesson introduced behavioral economics and explained in what ways it is different from traditional finance. We also covered how the transition from one to the other implies leaves behind a strict notion of utility maximization in favor of the new concepts of satisfice and prospect theory.

After establishing the need to bring individuals' behaviors into the picture, we analyzed the most typical cognitive and emotional biases displayed by individuals that are responsible for the need to change the way we look at finance to some degree.

The next lesson will cover in more detail the elements of prospect theory, explaining how prospect theory is applied and some of the mathematical tools on which it is based.

References

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Lesson Notes

MODULE 4 | LESSON 2

PROSPECT THEORY

Reading Time	90 minutes
Prior Knowledge	Students are expected to be familiar with the contents of Lesson 1 of Module 4 in which behavioral finance is introduced and the purpose of prospect theory highlighted.
Keywords	Value function and weighting function. Preference for skewness, Bowman's paradox, myopic loss aversion.

The previous lesson ushered us in the wold of behavioral finance and showed several key differences with respect to traditional finance. Prospect theory is the key tool for decision making, and an introduction to it was provided.

In the next lesson, we will apply the tenets of behavioral finance and prospect theory to portfolio theory.

1. Prospect Theory in a Formal Manner

In the previous lesson, behavioral finance was ushered in by assuming bounded rationality. Further, we introduced the concept of prospect theory in which risk aversion is replaced by loss aversion since Kahneman and Tversky, based on experimental findings, taught us that economic agents fear losses more than they value gains.

Now, the question becomes the following: Is prospect theory useful for analyzing investment decisions, i.e., decisions that deal with risky outcomes?

To answer this question, we need to understand how prospect theory works when individuals need to come up with decisions/choices.

Prospect theory consists of two phases:

- **Editing phase:** aimed at reformulating and/or reorganizing the available options of the problem to the end of simplifying the decision making process. It is implemented via several operations, which will be explained below.

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- **Evaluation phase:** the decision maker is expected to evaluate each of the edited options and choose the prospect with the higher value. As explained by Kahneman and Tversky, the overall values of the edited prospects, which we choose to denote by V, are expressed in terms of two scales. The first scale associates to each physical probability, p, a decision weight, $w(p)$. The second scale v assigns to each outcome x a number $v(x)$, which can be regarded as a subjective value of the outcome x. In particular, we stress that the outcomes are defined relative to a reference point, and thus, V measures the values of deviations from the reference point or, in other words, the value of specific wins or losses. We have already studied this part in the previous lesson.

1.1 Editing Phase

The editing phase involves several steps or, more properly, operations that transform the outcomes and the probabilities related to the prospects. The steps are:

- **Coding:** each proposal is coded as a gain or a loss of value relative to a reference point. Usually, but it is not strictly necessary, the reference point corresponds to the current value of the asset that is involved. Furthermore, probabilities are assigned to each possible event. Note that the values and losses can be different for different individuals.
- **Combination:** used to simplify the prospects by combining the probabilities of outcomes with identical values.
- **Segregation:** aimed at separating expected returns into risky and risk-free components (if possible, of course).
- **Cancellation:** removing any outcomes common to two or more proposals does not affect the decision.
- **Simplification:** described as the process consisting in rounding the probabilities of the outcomes of a prospect in the sense that if they differ by very small amounts or the outcomes are very unlikely, we can consider the probabilities the same or zero, respectively.
- **Detection of dominance:** we could discard from consideration any alternative that is clearly dominated and eliminate it from any further evaluation.

Examples are necessary to illustrate each of these entries and make them clearer.

Example 1. Consider a portfolio that consists of four assets: A, B, C, and D. Let's assume that we have the following table of information:

Asset	Current price	Historical price (cost basis)	Close price, yesterday
A	10	8	11
B	13	13	14
C	15	12	16
D	19	12	20

And suppose that we want to detect which asset suffered the largest percentage loss. To do this, we need first to decide which reference points to use: Is the reference point yesterday's close, or is it cost basis? Or perhaps we should use the year-end closing prices? If we used the cost basis, then the percentage change in prices would be:

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$$+25\%, 0\%, 25\%, 58.33\%$$

in which case the asset with the largest percentage loss would be B with 0%. If we used yesterday's close, then the percent changes in price would become:

$$-9.09\%, -7.14\%, -6.25\%, -5\%$$

and we would select asset A. Hence, a reference point is necessary, but what we performed after that is what Kahneman and Tversky described as coding.

Example 2. Again, we describe the situation with a table:

Outcomes	probability of outcomes	Expected return for that outcome
x1	0.09	-10%
x2	0.15	15%
x3	0.15	8%
x4	0.25	15%
x5	0.15	15%
x6	0.21	20%

This table displays three outcomes as having the same expected value (0.15), and they are x2, x4, and x5. Using combination, this table can be simplified as follows:

Outcomes	probability of outcomes	Expected return for that outcome
x1	0.09	-10%
x3	0.15	8%
x2, x4, x5	0.55	15%
x6	0.21	20%

So, combination has allowed us to simplify our table.

Example 3. Suppose that a gamble offers the following payoffs:

outcome	probability	payoff
x1	0.60	$100
x2	0.40	$130

This gamble is the same as the following:

outcome	probability	payoff
x3	1.00	$100
x4	0.40	$30

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In other words, by using segregation, we were able to reduce the original gamble to a new one in which the gambler earns $100 with probability 1 (riskless) and $30 with probability 0.40.

Example 4. Consider two proposals and their expected values and probabilities as described in the following table:

Proposal A		
outcome	probability	payoff
x1	1/3	10
x2	1/3	20
x3	1/3	30

Proposal B		
outcome	probability	payoff
x1	1/2	10
x2	0	20
x3	1/2	30

If we use cancellation, the outcomes that are common to both proposals can be removed and we would get the following tables instead:

Proposal A		
outcome	probability	payoff
x1		
x2	1/3	20
x3		

Proposal B		
outcome	probability	payoff
x1	1/6	10
x2		
x3	1/6	30

which is simpler to use.

Example 5. Let's say that a gamble has the following payoffs and probabilities:

outcome	**probability**	**payoff**
x1	0.49	$500
x2	0.50	$800
	0.01	$1000

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Simplification can be used, and it yields the following table instead:

outcome	probability	payoff
x1	0.50	$500
x2	0.50	$800
x3	0.00	$1000

Example 6. Detection of dominance is quite simple. Consider two proposals, A and B, whose payoffs and probabilities are summarized by the following table:

Proposal A		**Proposal B**		
outcome	probability	payoff	probability	payoff
x1	0.2	10	0.2	5
x2	0.5	20	0.5	15
x3	0.3	30	0.3	26

Since for each outcome the payoff of proposal A is higher than the equivalent payoff of proposal B and the probabilities of the outcomes are the same we can say that proposal A dominates proposal B (higher average, higher minimum, higher maximum). Hence, proposal B can be discarded.

1.2 Evaluation Phase

In lesson 1 of this module, we have already seen how to evaluate prospects. Here, we will be a bit more formal. Once the editing phase is completed, the decision maker is expected to evaluate each of the edited prospects and choose the prospect with the highest value. Kahneman and Tversky (299) explain "loss aversion, risk seeking, and nonlinear preferences in terms of the value and the weighting functions."

Hence, if we denote by V the overall value of an edited prospect, V can be expressed in terms of two scales: w and v.

- The first scale associates to each physical probability, p, a decision weight $w(p)$ which reflects the effect of the probability on the total value of the prospect.
- The scale v assigns a number to each outcome x, $v(x)$. The value v is subjective, and it measures the value of deviations (gains or losses) from the reference point.

Thus, given a set of possible outcomes

$$V(x_1, \ldots, x_n) = p_1 \cdot w(p_1) \cdot v(x_1) + p_2 \cdot w(p_2) \cdot v(x_2) + \ldots + p_n \cdot w)p_n) \cdot v(x_n).$$

the "utility" function in prospect theory looks like:

$$V(x_1, \ldots, x_n) = w(p_1) \cdot v(x_1) + w(p_2) \cdot v(x_2) + \ldots + w(p_n) \cdot v(x_n).$$

Example 7. Suppose that we are evaluating a prospect. Without loss of generality, We can assume that the lowest probability weighted payoff is achieved for the outcome x_1 (if that is not the case, w always relabel the outcomes so that the outcome with the lowest weighted payoff

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$w(p_i) \cdot v(x_i)$ is achieved for $i = 1$.). Then, during the editing phase, segregation allows us to write.

$$V(x_1, \ldots, x_n) = w(p_1)v(x_1) + \sum_{i=2}^{n} p_i[(w)p_i(v(x_i) - w(p_1)v(x_1)]$$

Hence, in this case, we have a riskless component $w(p_1) \cdot v(x_1)$ which is the minimum gain or loss and a risky component that depends on the other outcomes and their probabilities.

We need to spend a bit more effort on the value function. We have justified an asymmetric S shape for it, but we can say more. For instance, Kahneman and Tversky in their 1992 paper proposed the following function

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0; \\ -\lambda(-x)^{\beta} & \text{if } x < 0 \end{cases}$$

and estimated the parameters as $\alpha, \beta \approx 0.88$ and $\lambda \approx 2.25$. The fact that $\alpha \approx \beta = 0.88$ informs us that there is 1.) a diminishing sensitivity to both positive and negative extreme outcomes, and 2.) substantial loss aversion since $\lambda = 2.25$ describes the slope of the loss function vs. the gain function. In practical terms, the disutility of giving something up is more than two times the utility of gaining the same amount.

Another approach exploits cumulative distribution functions. In particular, using this approach, we can obtain a parametric version of the value function. This is useful as we can adjust it to different instances. To see how this works, let Y be a random variable that takes only nonnegative values and assume that $Z = \log(Y)$ follows a logistic distribution. Z is known as a log-logistic distribution. This means that Z has a CDF that looks like this

$$F(z, \mu, \sigma) = \frac{1}{1 + (z/\mu)^{-\sigma}}.$$

By multiplying both numerator and denominator by $(x/\mu)^{\sigma}$ and some other algebraic manipulations, it is possible to rewrite

$$F(z, \mu, \sigma) = \frac{z^{\sigma}}{\mu^{\sigma} + z^{\sigma}}.$$

Now, by changing the values of the parameters μ and σ, we can obtain many different versions of the value function. This function is not completely satisfactory for our purposes yet. However, if we shift it left and down, then we obtain a veritable candidate for the value function:

$$F(z, \mu, \sigma) = \frac{(z + \mu)^{\sigma}}{\mu^{\sigma} + (z + \mu)^{\sigma}} - \frac{1}{2}.$$

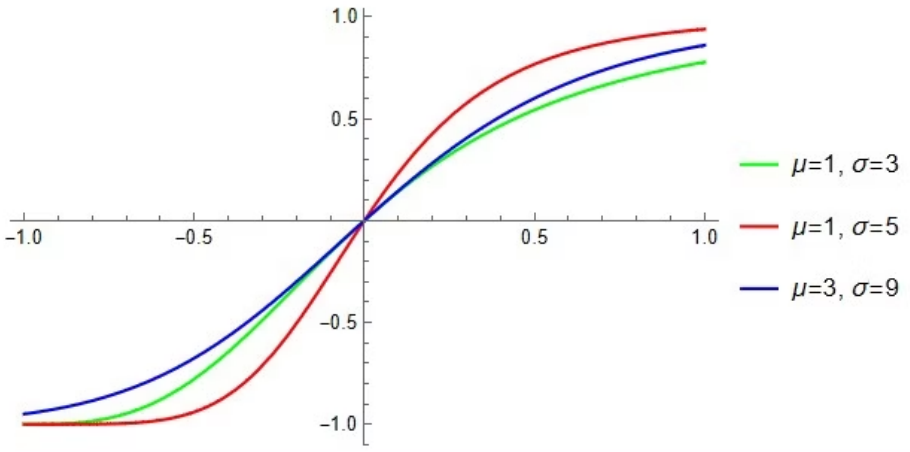
Finally if we multiply it by 2, we get a v function with asymptotes ± 1 :

$$F(z, \mu, \sigma) = \frac{2(z + \mu)^{\sigma}}{\mu^{\sigma} + (z + \mu)^{\sigma}} - 1.$$

In [illegible]e 1, three such value functions have been plotted to see how the choice of the parameters μ a[illegible]ffect the overall shape of the value curve.

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Figure 1: Value Function Derived from the Log-Logistic Distribution



Another suitable candidate for the value function can be obtained using a lognormal distribution. Although it is not possible to write the CDF of the lognormal distribution in a closed form fashion, it can nevertheless be expressed as

$$F(\ln(x),\alpha,\beta)=\frac{1}{2}+\frac{1}{2}\mathrm{erf}\left(\frac{\ln(x)-\alpha}{\beta\sqrt{2}}\right).$$

with erf denoting the error function, i.e.,

$$\mathrm{erf}(x)=\frac{2}{\sqrt{\pi}}\int_0^x e^{-u^2}du.$$

The log-logistic and the log-normal distributions, once the parameters α,β,μ,σ are chosen in a suitable manner, produce very similar value functions and are both very suitable to produce value functions that can be adapted to different real-life situations using mathematical models with just two parameters.

Now, we need to spend a few words about the probability weight function. In the previous lesson, we covered Kahneman and Tversky's motivations. Earlier versions used monotonic transformation of outcomes probability and that did not always produce the desired conclusions theorists had in mind. Kahneman and Tversky's original contribution is seen in the fact that they use a "rank dependent functional" in which one does not transform each probability separately, but the entire cumulative distribution function. In the previous lesson we considered situations in which the random outcome were just a few; however it would be difficult to implement that approach if we had tens and tens of outcomes and even more so if we had a continuous of outcomes.

Empirical studies suggested an inverse S-shaped weighting function, concave below a certain threshold and convex above the same (to be consistent with risk-seeking for low probability gains and risk aversion for high probability gains). Kahneman and Tversky proposed a family of weighting functions that uses two parameters:

$$w^+(p)=\frac{p^\gamma}{[p^\gamma+(1-p)^\gamma]^{1/\gamma}}$$

for the probability of outcomes that lead to gains, and

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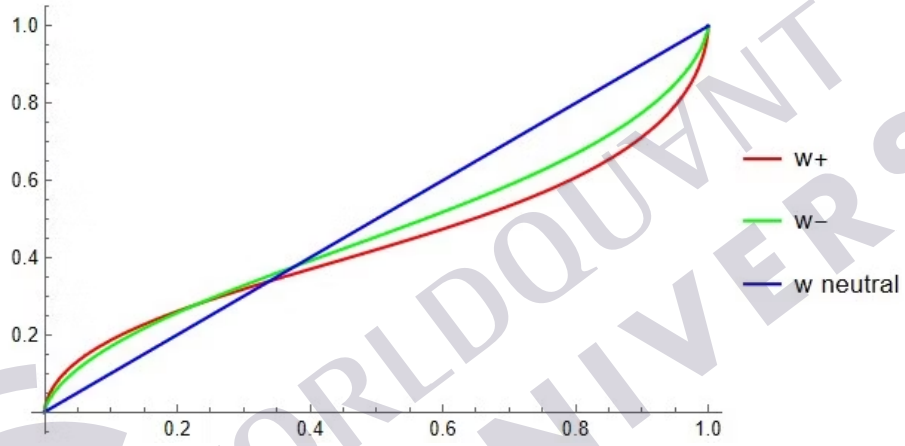
$$w^-(p) = \frac{p^\delta}{[p^\delta + (1-p)^\delta]^{1/\delta}}$$

for the probability of outcomes that lead to losses instead.

By modifying the parameters γ and δ, we can change the "stretching" of the weighting function. The curvature of the weighting function aims at depicting the individuals' attitude towards low, moderate, and high probabilities. In particular, when $\gamma = \delta = 1$, then we get back the original probabilities, i.e., $w^{\pm}(p) = p$, which we could call the perfectly risk-neutral weighting. In the same 1992 papers we mentioned earlier, Kahneman and Tversky estimated $\gamma = 0.61$ and $\delta = 0.69$.

This particular weighting function is displayed in figure 2. It is clear from the picture that individuals underweight high probabilities both in the case of gains and losses, but more so in the case of gains. Similarly, individuals overweight small probabilities for both gains and losses, but slightly more so in the case of losses.

Figure 2: Kahneman and Tversky Weighting Function



A study by Gurevich et al. using stock option data for the U.S. produced values of $\gamma = 0.84$ and $\delta = 0.76$. This agrees with the result of Kahneman and Tversky, although in this case, the w^+ and w^- transformations are closer to the straight line of risk-neutral weighting.

Remark: The paper by Gurevich et al., while short, does not make for a truly simple reading as it requires the knowledge of stochastic discount factors and state prices, which are usually studied in advanced classes in financial economics. Nevertheless, it is highly recommended for students in quantitative fields. Behavioral finance does not use complex optimization calculations like traditional finance; however, it requires a certain mix of quantitative abilities like estimating the right value and weighting functions. For that reason, the reference to this paper is provided, and it makes for a useful voluntary reading. The mathematics used in the paper is not particularly advanced.

In what follows, we will use prospect theory to explain certain paradoxes or puzzles. This highlights that prospect theory is not just an academic creation for show, but it can actually explain real human behaviors as far as investing is concerned.

2. Preference for Skewness

Individual investors can be optimistic or pessimistic. In some ways, we could say that a positive s ent for the market leads investors to expect optimistic outcomes while a negative sentiment t a pessimistic sentiment about the markets, which leads to expecting negative outcomes.

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However, the evidence suggests that among individual investors, the difference between pessimistic or optimistic individuals does not lie in how they allocate their portfolios to equities, bonds, and cash, but rather in the amount of diversification that they employ in allocating their portfolios. More specifically, optimists tend to ignore the benefits of diversification—so much so that some authors have asked themselves why optimism tends to have a positive effect on the rate of success of entrepreneurs and self-employed people in general. Nevertheless, it fails to produce a similar outcome among investors where the results are usually not good, if not outright deleterious. In this sense, Statman (44) noticed in a 2004 paper that there is a problem (diversification puzzle):

"The levels of diversification in U.S. investors' equity portfolios present a puzzle. Today's optimal level of diversification, measured by the rules of mean variance portfolio theory, exceeds 300 stocks, but the average investor holds only 3 or 4 stocks."

This happens because:

1. optimists overestimate returns or underestimate risks;
2. pessimists underestimate returns or overestimate risks.

These facts can be revisited in more mathematical terms by stating that optimists have a **preference for skewness**; i.e., they are willing to accept lower returns in exchange for positive skewness and the possibility of very large payouts. This preference for skewness can be incorporated by modifying the weighting function.

Earlier, we used the following piecewise function introduced by Kahneman and Tversky:

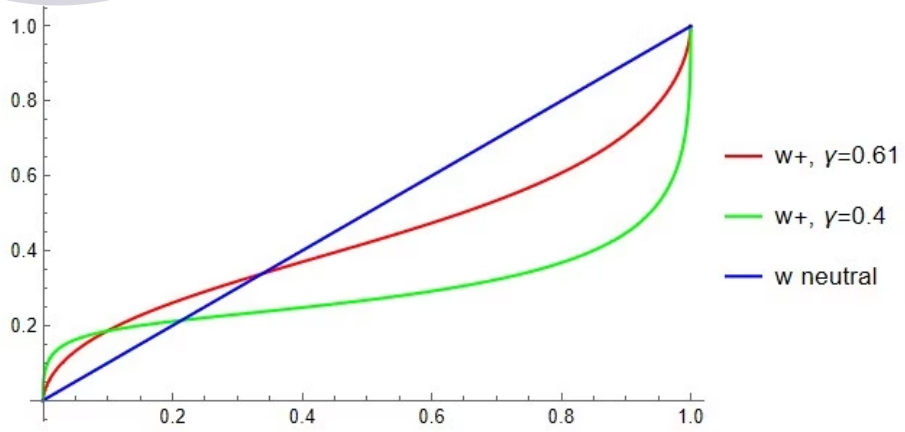
$$w^+(p) = \frac{p^\gamma}{[p^\gamma + (1-p)^\gamma]^{1/\gamma}}$$

for the probability of outcomes that lead to gains, and

$$w^-(p) = \frac{p^\delta}{[p^\delta + (1-p)^\delta]^{1/\delta}}$$

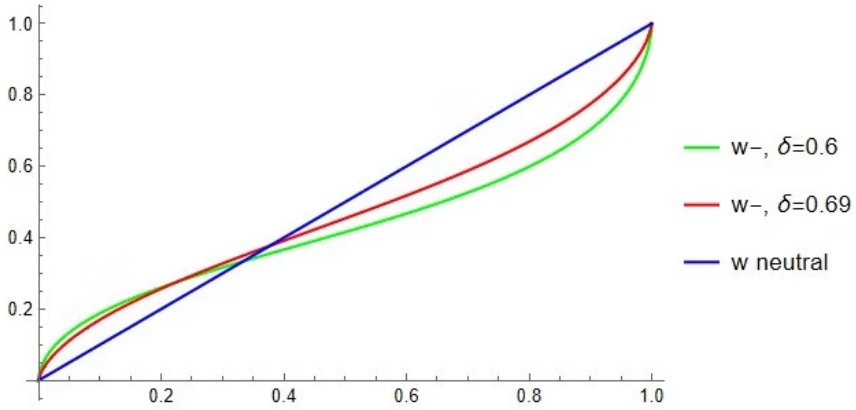
for the probability of outcomes that lead to losses, instead with $\gamma = 0.61$ and $\delta = 0.69$. The preference for skewness can be modeled by using a different set of parameters, say $\gamma = 0.4$ and $\delta = 0.6$ (see figures 3 and 4).

Figure 3: Preference for Skewness: Weighting Function for Gains



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Figure 4: Preference for Skewness: Weighting Function for Losses



In these figures, it is clear that there is a more extreme over-weighting of low probabilities and under-weighting of large probabilities than in the original function used by Kahneman and Tversky.

The relationship we just highlighted between low returns, positive skewness, and investors' preference toward investments that offer lottery-like payoffs is observed across a wide variety of financial settings. One recent trend that witnesses this type of behavior is the increasing participation in lotteries and the like.

3. Bowman's Paradox Explained via Prospect Theory

William J. Bernstein—a neurologist who later became a financial analyst—pointed out that in spite of the general expectation that growth stocks will earn a higher return than value stocks due to the fact that they are usually high beta stocks and thus contain a lot of systematic risk, which makes them overall riskier than value stocks, in reality, historical returns point to the opposite. The fact that low volatility stocks earn higher returns than high volatility stocks poses a serious challenge to the notion of rational expectations and has deserved its own name of volatility anomaly.

Edward H. Bowman observed in 1980 that the riskiest firms provided the lowest returns, a result confirmed in further articles in 1982 and 1984. The difference with what we discussed in the previous paragraph is that Bowman uses a *firm-level accounting-based risk and return and not financial returns.* More specifically, Bowman uses mean values of ROA as a proxy for return and variance of ROA as a proxy for risk. His approach has been criticized on the basis that using moments of the same variable might lead a statistical bias since the moments of any same variable could be positively related to each other. Furthermore, successive studies reported that skewness of return distributions can lead to spurious negative relationships between risk and return. Nevertheless, further studies have not been able to invalidate these findings, not completely in any case. And, in at least one instance, when total debt to total asset ratio is used as a risk measure, significant negative association between performance (measured via ROA or ROE) and leverage for all company size categories were detected. This has become known as **Bowman's paradox** and presents a challenge to the strategic management's tenets that higher risk needs to be compensated by higher returns; otherwise, management would not be motivated to undertake risky investments/projects. In fact, Bowman observed something even stronger, i.e., that high levels of managerial skills produced low returns more often than high returns.

Several attempts have been made to explain Bowman's paradox in the context of TF. Prospect t[illegible]has been invoked as well as using the facts that individuals use reference points or targets to

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evaluate risky choices and individuals are not risk-averse in a uniform manner. In fact, when returns have been below the target, managers can become risk-seekers, and instead, when returns have been above a given target, the majority of managers become risk-averse. In other words, companies may be prone to behave like individuals. Note that the reference point plays a very important role in this formulation. What is the reference point? In this case, it is a suitable target level of return, which could be based on industry-wide averages or some industry-wide benchmark.

Applications of prospect theory to solving the Bowman's paradox have revealed that the expected relationship between risk and return fails to materialize for firms with returns close to the target level—or that in some cases, there is no significant correlation between the two around the threshold level of the return target.

4. Equity Premium Puzzle and Market Participation Puzzle: A Lesson from Prospect Theory

Another situation that has puzzled economists for a while has to do with the equity risk premium. In finance, we are taught that investors do not like risk and demand higher returns to bear it. The problem that several econometric works have shown is that the size of the equity risk premium is too high to be rationally justified. Despite this higher than rationally justifiable premium, there is a strong preference among investors for bonds vs. equity. Even considering indirect holdings (via pension funds or IRAs), only about 37% of American households hold stocks. And this has not changed even after the defined benefit pension plans have started to fall out of use. The Federal Reserve in a 2014 survey reported that the total participation in the stock market (direct and indirect) had fallen from 2007 to 2013. The Federal Reserve estimated that the overall participation is 48.8% and only about 14% of American households hold stocks directly.

This has created a **stock market participation puzzle,** i.e, why are American households failing to take advantage of the equity risk premium and the usually low correlation with other household risks? And it is not possible to dismiss the puzzle based on education or wealth, for other studies have highlighted that the participation in equities is even lower among wealthier investors (80th percentile of wealth or higher) at only around 10%.

To make all this more surprising still is the fact that there have been clear warnings about individuals' lack of preparedness for retirement in recent times.

Behavioral finance has pointed out that investors have a tendency to check the performance of their portfolios too frequently. With equity investments, doing this causes them to experience many days of losses. If they were to check their investments' performance over longer periods of time, losses would be less frequent. Now, prospect theory teaches us that investors feel the pain of losses far greater than they feel the pleasure of gains. Therefore, the frequent checking of results is a cause of pain and even panic.

Behaviorists have created the term **myopic loss aversion** to describe this combination of short time horizon and framing bias. How can myopic loss aversion impact investment results? Well, investors who check on the values of their portfolios with great frequency are more likely to suffer from this particular bias. And, this is arguably even more so in the internet age as investors are able to check their portfolios in real time or nearly in real time and are thus subjecting themselves to the pain and p[illegible]f losses. The end result is that, reacting to these frequent losses, their behavior is to stray

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from sound investment plans, and they are even more likely to do so in bear markets where the frequency in experiencing losses can make investors very susceptible to frequent trades with a high risk of ending up buying high and selling low, which is the opposite of ideal.

This explanation can also serve to justify the existence of the equity risk premium puzzle: Myopic loss aversion causes investors to demand higher risk premiums to hold on to stocks that could indeed go up but also down, causing said investors to experience pain. In this instance, the functional forms of the value function or the weighting function are not the critical elements; the key factor at work is some sort of psychological cost that stems from the fact that individuals are using the wrong time horizon to evaluate their investments and that pushes them to behave in the way they do.

5. Conclusion

The lesson analyzed prospect theory in details by presenting the two different main phases of editing and evaluation. Then, prospect theory is used to explain a few paradoxes or puzzles that trouble traditional finance.

Once we accept that traditional finance and assuming that economic agents are completely rational do not represent the behavior of real people, the question becomes how can they achieve their financial goals? If an investor displays behavioral biases and uses prospect theory instead of being fully rational in the sense of traditional finance, how do we help them? Behavioral finance suggests that we embrace a mix of adapting and moderating the investor's biases as that would ultimately lead to a better overall result than telling them how to invest. The way to do this is called behaviorally modified asset allocation (BMAA), and it will be covered in the next lesson.

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Lesson Notes

MODULE 4 | LESSON 3

BEHAVIORAL PORTFOLIO THEORY

Reading Time	60 minutes
Prior Knowledge	Lesson 1 and Lesson 2 of Module 4.
Keywords	Behavioral Portfolio Theory (BPT), Adaptive Market Hypothesis (AMH), relative wealth, standard of living risk, Behaviorally Modified Asset Allocation (BMAA)

The previous lesson dealt with the formal aspects of prospect theory and how it can be applied to real-life situations. In particular, we investigated how prospect theory is capable of explaining certain puzzles and anomalies that are difficult or impossible to justify within the realm of traditional finance.

In this lesson, we look at portfolio theory. Traditional finance with its formulas and perfectly rational individuals is just too difficult for many individuals to accept, and this is true in particular when investors' portfolios suffer losses. Behavioral portfolio theory and behaviorally modified asset allocation are the tools proposed by behaviorists to better serve investors and ultimately assist them in reaching their financial goals.

1. Problems with Traditional Portfolio Theory

Up to this point, we have learned more than a little bit about how to construct optimal portfolios. However, all we did was based on requiring that individuals act as **rational economic men (REM)** who are able to access all relevant information and capable of using new information to upgrade their solutions. We also insisted on them being risk-averse. The first two lessons of this module, on the other hand, argued that human beings do not exactly behave like the theory tells us they should.

TF makes one key assumption: **Markets are efficient** and prices reflect the fundamental value of assets. Furthermore, TF endorses the notion that, as soon as new information becomes available, it is quickly and correctly transferred to market prices. The implication of these two postulates is that managers can focus on identifying which portfolios on the efficient frontier are compatible with their clients' objectives in terms of risk and return. Nothing else seems to matter. We could discuss whether markets are truly strong efficient or semi-strong efficient, or even just weak efficient.

However, when prices no longer represent the correct intrinsic value of assets, the traditional approach to portfolio theory is flawed and not just in a minor way.

We must also notice one further aspect of TF: It assumes implicitly that investors, in going through different stages in their lives, are able to save and invest in the early stages to fund their expenses in retirement. While on paper this seems a sensible behavior to expect from rational investors, it requires that investors are capable of self-control, i.e., they are capable of delaying short-term spending and gratification in exchange for achieving long-term goals. Unfortunately, there have been plenty of experiments that display that actual individual behaviors run contrary to self-control. If that were not enough already, in many situations, even the very process of thinking in terms of saving now to consume later represents a challenge since individuals are prone to suffer from biases like mental accounting and framing. For instance, it has been observed that individuals treat the money they earn as wages differently from the money they earn as bonuses or by winning a lottery: If they perceive a bonus as part of their regular compensation, they are more likely to spend it while if they perceive it as exceptional instead, they are less likely to spend it and may save a bigger percentage of it.

Consider again the equity premium puzzle discussed in the previous lesson. TF is quite clear in that the value asset is calculated by considering the expected cash flows over the life of the investments and discounting them using a rate that depends on the risk-free rate and some suitable risk premium. This requires that one knows the correct discount rate to use. BF, on the other hand, adds a *sentiment premium* to the discount factor. Roughly speaking, this sentiment premium can usually be estimated by considering the dispersion of the analysts' forecasts where a high dispersion implies a high sentiment premium (more dispersion = more risk). In TF, there is no need for a sentiment premium, and besides, if it were added and predictable, it would be, or at least it could be, exploited and eventually eliminated, possibly via arbitrage. The presence of a sentiment risk premium can change the perception of risk and reward among individuals, and it raises the question of whether sentiment can be considered a priced source of risk or if it causes mispricing instead. (Readers interested on these issues should read the paper by Shefrin and Statman listed in the References section).

These are serious issues that should be addressed.

2. Behavioral Portfolio Theory (BPT)

At the risk, perhaps, of repeating ourselves, though an old saying in Latin insists that *repetita iuvant*, i.e., to repeat is beneficial, we recall that in TF, we end up with a diversified portfolio that satisfies the investor's preference for risk and return by selecting possible combinations of assets from the set of mean-variance efficient portfolios (the efficient frontier). That is the sensible thing to do as TF is built upon the tenet that investors are concerned about expected return and risk of the whole portfolio, i.e., a combination of assets that is considered in its totality.

Optimal portfolios in TF can be regarded as the optimal portfolios of CAPM investors, and as we saw in Module 1, they combine a market portfolio with a risk-free asset.

Optimal portfolios of BPT investors resemble combinations of bonds and lottery tickets instead. This happens because BPT is based on a very different principle: Investors build their portfolios in layers, and for each layer, there is a different risk and return preference. Hence, investors do not compute an overall optimal portfolio, but an optimal portfolio for each layer. This is well explained in

the 2000 paper by Shefrin and Statman, a reading that is not very technical and strongly recommended to students who are interested about the behavioral perspective.

According to BPT, portfolios are built to maximize wealth with one important constraint: Given an investor's most important goals, we want a low probability of failing to meet these objectives. Assets are separated in layers based on risk. The low-risk layers will contain bonds or riskless investments; the high-risk level will consist of certain speculative equities, private equity investments, derivatives, etc. This version is called a **multiple mental account version** (**BPT-MA**) by Shefrin and Statman. There exists also a a single mental account BPT version (**BPT-SA**) in which investors, like mean-variance investors, integrate their portfolios into a single mental account by considering covariance among the different layers. However, this version is less popular, and when we refer to BPT, we will do so meaning BPT-MA.

Each investor has certain goals in mind: Meeting basic needs, paying for children's schooling or college in the future, providing for a sick parent in their old age, buying a fancy car, buying a vacation home, etc. Some of these goals will be deemed essential; others can be considered just desirable. For instance, providing for basic needs or for a sick member of the family would be considered essential, and therefore, the investor has little or no desire for speculative investments as keeping the risk low is the priority. Buying a vacation home in Aspen, Colorado, may be very desirable, but it is not essential. In this case, the investors may be willing to accept riskier investments to get closer to this goal but would not produce dramatic consequences if the required return is not achieved.

With all this in mind, the process of constructing portfolios according to the behaviorists' perspective goes through the following steps:

- Each layer's fund allocation is determined by the investor's goals. If the goal requires a high return, funds will be allocated to a high return/high risk layer. On the other hand, if the goals require a low risk, then financial resources will be allocated to the low return/low risk layer.
- Asset selection will be done on a layer by layer basis.
- In each layer, the number of assets will reflect the investor's risk aversion. In practical terms this means that if an investor is risk averse (concave utility curve), they will hold a more diversified portfolio involving a larger number of assets than an investor whose utility curve is convex or concave and convex. This is applied on a layer by layer basis.
- In the case an investor comes across some information that is not public knowledge, they can tweak the portfolio composition toward more concentrated positions.
- When investors are loss-averse, they will hold larger cash positions to avoid liquidity requirements forcing them to sell assets at a loss. In addition, there could be instances where an investor holds on to certain assets even if their potential performance is not attractive simply because they want to avoid realizing losses.

The optimal portfolio that comes out of this process could give the impression of being diversified, but since the layers were constructed without considering the correlation among the different layers, it is likely a sub-optimal combination. The reason BTP-MA is relevant is that it is able to explain what one observes in real life:

- individuals holding both low risk assets and high risk assets
- holding excess cash in low risk layers and excessive risky assets in high risk layers and only modest amounts of moderately risky assets.

3. From Efficient Market Hypothesis to Adaptive Market Hypothesis

The concept of the adaptive market hypothesis (AMH) originated in a 2004 paper by Andrew Lo and can be summarized with the formula:

$$\text{AMH} = \text{EMH} + \text{bounded rationality} + \text{satisficing} + \text{"evolution"}.$$

In essence, AMH assumes that market participants, to be successful, apply heuristics instead of hard core optimization. In addition, they do so as long as these heuristics work, and once they stop working, participants modify them and/or try to come up with new ones.

This behavior suggests that **winning strategies are an evolutionary game**. Nevertheless, there is not just a change in how the "game" is played, but a change in the goals as well: Unlike TF where the goal is maximizing expected utility, now the objective is to survive in competitive markets, and those who cannot adapt may not be able to survive.

The presence of satisfice should be recognizable from the fact that in the previous paragraph we acknowledged the fact that market participants are not performing a hard core optimization, but based on the amount of information that they consider sufficient, they make decisions to reach sub-goals on their path towards achieving their ultimate goal. The use of heuristics means that economic agents are able to come up with decisions via trial and error, and these decisions, unlike theorems, are not supposed to work all the times. Nevertheless, investors keep using and updating them until, at a certain point, their rules are reflected in market pricing and no longer work.

AMH produces several practical consequences:

1. The relationship between risk and return is not stable in response to competitive changes;
2. Active management makes sense since it is possible to find arbitrage opportunities to exploit;
3. Strategies do not work all the time;
4. To be successful and stay successful, one has to adapt and innovate, and those who are best at changing and innovating will survive. The others may not.

We could ask ourselves one question: why are many investors ultimately not able to construct portfolios that are optimal in the sense of TF? Arguably, the most fitting answer is that they do not understand them and when investors do not understand them, they are not likely to follow TF's principles in the long run. What if they gave the money to a portfolio manager who is skilled and understands TF? That could be a solution; however, the experience in the field is that these investors would be happy as long as their portfolios do well but are likely to complain or even withdraw their money from the fund when things do not go the way they envisioned. Consequently, they are more than willing to blame the portfolio manager (or themselves for listening to their advisors) for any unfavorable outcome. The self-control bias can be applied even to how investors behave towards portfolio managers because, at a personal level or in the relationship with a portfolio manager, they could lack the necessary discipline to be TF investors and follow a path they do not understand or, at least, do not fully understand.

4. Behaviorally Modified Asset Allocation

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Behaviorally modified asset allocation, (BMAA) is an approach to asset allocation in which clients' behavioral biases are taken into consideration and incorporated into the allocations themselves.

What is the advantage of doing so? It serves to help financial advisors or portfolio managers to build portfolios for their clients in such a way that clients will not stray from them when they encounter periods during which their portfolios, for whatever reason, are under-performing compared to their expectations.

The last point is very important: If a client is upset by the fact that their portfolio lost, say, 10% over a period of a few months, they may abandon an otherwise sound strategy. Worse still, they may do so at a low point and, to add insult to injury, do it right before their portfolio begins to recover. Of course, this assumes that the poor outcome is only temporary.

For financial advisors, it is hard to advise their clients if their clients are not listening. BMAA is meant to help the relationship between financial advisor and client.

The ultimate goal of BMAA is to determine a strategy in advance that a client is more likely to stick to even during adverse periods.

A BMAA is created in several steps:

1. determine a strategic asset allocation that is optimal and consistent with TF;
2. evaluate the client's biases and, possibly, determine if they are mainly cognitive or emotional.
3. take into consideration the level of wealth of the client versus their lifestyle and needs/desires.

The first two steps are self-explanatory and do not require any further comments. The third step is perhaps less evident, and it can be better understood in terms of determining the size of the **standard of living risk**.

To illustrate this concept, considers two clients, both living a fairly lavish lifestyle, and let's call them Client A and Client B. Now:

- Client A has a financial portfolio worth $10 million, an annual salary of $500,000, and no children or other family members who require his assistance.
- Client B instead has a financial portfolio of $1 million, an annual salary of $250,000, and two children who are about to go to college for which he would like to pay. Additionally, he is taking care of his brother who suffers from a severe illness, is unable to work, and needs frequent expensive medical care that costs about $100,000 a year.

While they are both reasonably wealthy, in terms of relative wealth, Client A is in a much better position to maintain their basic lifestyle, and in particular, there is a very low risk that he might be unable to fulfill his essential needs. On the contrary, Client B is almost "living on edge."

Now, the concept of essential needs is not synonymous with "just enough to survive." Some individuals may think of lavish lifestyles as necessary to enjoy life at its fullest. Other individuals, on the opposite side of the scale, could think of essential needs as just putting food on the table and a roof over their heads. In some cases, it could mean being able to fulfill basic needs and being able to care for a sick member of the family.

What we are stressing here is that the definition of essential needs is personal to the particular client. A financial advisor will have to be able to recognize which is what.

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The point is that Client A has a low standard of living risk and could navigate troubled markets without much risk of being forced to give up his lifestyle. On the other hand, Client B has a high standard of living risk since even a moderate economic downturn could jeopardize his ability to meet his basic needs and obligations.

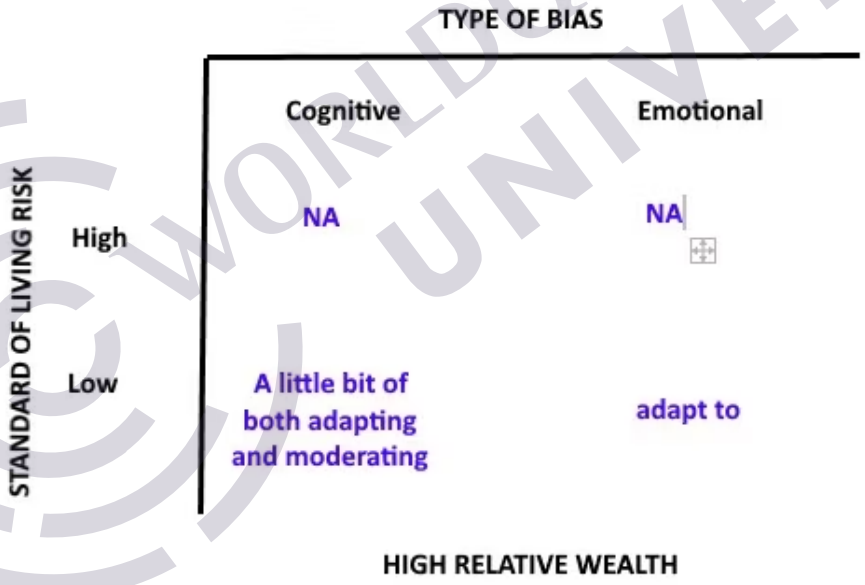
Having stressed this point, to come up with a BMAA portfolio, three initial determinations are required:

1. The level of relative wealth vs. financial goals. To facilitate the process, financial goals can be divided in three main groups: (a) basic needs (food, rent or mortgage, taking care of family, a specific lifestyle that is considered very important for the person), (b) priorities and desires (donating to charity, establishing endowments), (c) aspirational (owning an exotic car, a second home on the French Riviera, or even something much less flashy as leaving a bequest at death);
2. The client's level of standard of living risk (high or low);
3. The types of biases that the client displays (cognitive or emotional).

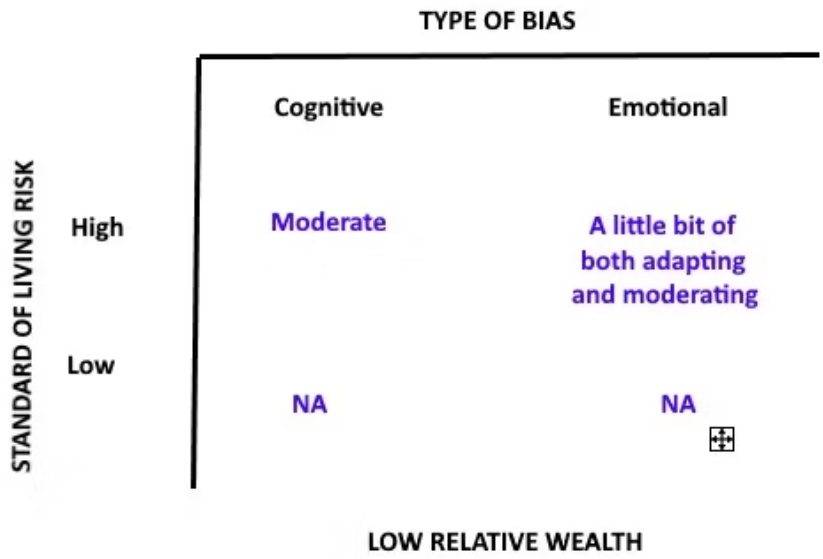
Then, we have to make a fourth and last determination:

1. Decide whether to mitigate/moderate or adapt to the client's biases. This is done according to a three-dimensional matrix (see figure 1):

Figure 1: BMAA at Work



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Let's try to understand the process of how the entries in the tables are chosen.

A high level of wealth vs. essential needs means a low standard of living risk. On the other hand, a low relative wealth in general means a high standard of living risk.

We have already explained that cognitive biases are easier to correct or at least moderate because it is possible to correct the faulty reasoning associated with them. On the other hand, emotional biases are more difficult to correct as the faults are deeply embedded in an individual's psyche. In the case of wealthy individuals with a low standard of living risk and emotional biases, we can indulge the client and adapt (within reason!) to their choices because they can afford to deviate more from an optimal portfolio then less affluent individuals. If we were dealing with a low relative wealth client with a high standard of living risk, regardless of whether the behavioral biases are emotional or cognitive, there is a need to introduce at least some moderation so that the client's portfolio does not stray too much from the optimal one.

What does it mean to adapt and to moderate in practical terms? There is no "one size fits all" sort of rule. However, we could think of something like this table:

Standard of living risk	Primary type of bias	Deviation (up or down) from optimal portfolio weights
Low	Emotional	15% - 20%
Low	Cognitive	5% - 8%
High	Emotional	5% -10%
High	Cognitive	< 5%

Note that the percent deviations listed in the table are just meant to favor a visualization of the relative importance of moderating but are otherwise arbitrary.

For a discussion about these issues, interested students should read the paper by Pompian and Longo listed among the References for this lesson.

Now, let's see where some of these ideas are put to work on an example that is similar to those encountered in real life—or, maybe, a bit dramatized for effect.

Example. You just started working for an important mutual fund in New York City, and John Gwinner is your first client. Following company guidelines, you conduct an extensive interview with Mr. Gwinner and have determined that:

- Mr. Gwinner has a portfolio of $1,000,000 and a salary of $180,000 a year, and he is expected to inherit a large sum of money from his mother's estate that he estimates at about $2,000,000. The company he works for does not provide a defined benefit pension plan.
- Mr. Gwinner plans to retire within the next three years.
- Mr. Gwinner explains that his main goal is to be able to use his financial portfolio and live a modest life after he retires and estimates that, to do that, he will need an income of approximately $75,000-85,000 a year.
- During the interview you also noticed that Mr. Gwinner displays clear tendencies toward conservatism, framing, representativeness, and endowment biases.

Should you adapt or moderate his portfolio?

Considering Mr. Gwinner's goals and current financial situation, his standard of living risk is low. The biases that were identified are mainly cognitive (the only exception being endowment), and thus, it is easier to moderate his biases. Therefore, according to the table above (which is fictional but useful for an example), we can accommodate a 5% to 8% deviation of his portfolio weights from the optimal values.

Three years later, you meet again with Mr. Gwinner for the annual interview with your clients. Several things have happened to him. His mother became severely ill and needs assisted care 24 hours a day. His son was in a bad car accident and is now unable to work. Mr. Gwinner would like to provide for his niece to attend a prestigious prep school as Mr. Gwinner's son planned. Furthermore, Mr. Gwinner's portfolio did not perform particularly well and is currently valued at $750,000. He is about to retire and his social security pension will be about $2,500 a month. His mother's estate now stands at about $500,000 due mainly to her health situation and medical bills. Mr. Gwinner is concerned that he would now need $150,000 a year, double the amount he envisioned three years ago. His behavioral biases have not changed.

How does Mr. Gwinner's new situation change your recommendations regarding his portfolio?

Mr. Gwinner is not in dire straits, but his mother's situation, his imminent retirement and consequent loss of income, together with his desire to pay for his niece's prestigious school, have changed his situation quite a bit, and there is a higher standard of living risk than 3 years ago (his needs are realistically felt as essential). This fact combined with his mostly cognitive biases would indicate that you should try to moderate his biases and allow only small deviations from the optimal weights for the assets in his portfolio (< 5%).

In the next lesson, we will cover a couple of formal behavioral models (the BB&K 5-way model and the Pompian model more in detail).

At this point, we may ask ourselves: Where did the fancy portfolio math of the previous modules go?

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Well, it is still there in the sense that one still needs to determine an optimal portfolio based on the client's preferences for risk/return. However, we have become amenable to accepting certain deviations from those optimal weights, and in some instances, we can be more accommodating than in others. Nevertheless, we are no longer telling clients that they must invest following nearly definitive rules.

Again, it is not possible to transform an individual who exhibits bounded rationality and behavioral biases into an REM. Doing so can make the client hostile and cause the individual to "rebel" against the portfolio manager's advice or even withdraw the money invested and bring it somewhere else.

5. Conclusion

This lesson explained how to build a financial portfolio according to behavioral finance. It covers the changes that are required to approach the problem with a behaviorist's point of view and introduces the concept of behaviorally modified asset allocations while providing a description of the key parts of it.

The next lesson will look at two well-known models to classify investors in order to assist financial advisors in understanding the personality of their clients in order to better serve them. We will also look at some important sub-optimal behaviors that affect investors and financial advisors/portfolio managers.

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Lesson Notes

MODULE 4 | LESSON 4

Behavioral Types: Behavioral Explanations for Financial Bubbles and Crashes Lesson Notes

Reading Time	90 minutes
Prior Knowledge	Module 4 Lesson 1, Lesson 2, and Lesson 3
Keywords	Barnewall's behavioral model, BB&K behavioral model, Pompian model, momentum, financial bubbles and crashes

In Lesson 3, we learned that behavioral portfolio theory requires that investors' behavioral biases be incorporated in the creation of portfolios. To this purpose, we identified the need to detect whether an individual's biases are mainly cognitive or emotional.

In this lesson, we will learn how to classify investors based on certain behavioral models, which are supposed to produce a better picture of our investors in terms of their ability to bear risk and how to improve the relationship between client and financial advisor/portfolio manager.

In the second part of the lesson, we will look at certain events like financial bubbles and crashes, which should not even exist according to Traditional Finance, and we will explain them from a behaviorist's point of view.

1. Classification of Investors

We have already stressed with sufficient emphasis in the third lesson of this module that, when the behavioral biases of investors are taken into proper consideration, the result is a favorable outcome for the investors.

In that lesson, we also emphasized the shift from traditional finance to behavioral portfolio theory. In TF, clients are *simply unemotional* and equipped with formulaic measures of return and risk. For instance, we could tell a customer that they are better off with a specific portfolio out of several available simply because that portfolio has a higher Sharpe ratio. However, if the client is an individual who reacts emotionally when discussing investments, it is quite likely that they will not find a metric like the Sharpe ratio of much relevance. However, it remains very important to enlist the cooperation of clients as, in the end, they are the only ones who can decide whether to follow our advice or go with their emotional feelings.

Sometimes, a bit of bending of strict TF principles produces much better overall results as clients feel engaged, become easier to advise, and are ultimately happier with their portfolios and more likely to follow strategies and avoid costly mistakes.

We have already discussed the concepts of adapting or moderating in Lesson 3, we also learned to think about the clients' goals and their risk of standard of living, although, mostly in the quizzes, we saw the need to interview customers to ascertain their specific behavioral types.

Behavioral finance has developed a few behavioral models to help identify customers.

1.1 Barnewall's Behavioral Model: Passive and Active Investors

This is a two-way model that dates back to 1987 (see the References section) and in which investors are classified either as passive or active.

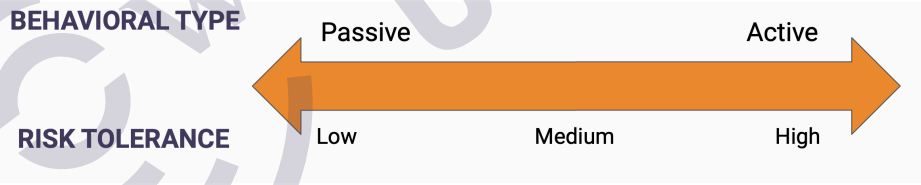
A **passive investor** can be described as an investor who has accumulated wealth through a long and steady employment, very disciplined saving, or by inheriting money.

An **active investor**, on the other hand, is someone who has accumulated wealth by taking an active role in investing their money.

Why is this simple classification important? It has to do with their attitude towards risk:

- Passive investors are more risk averse and are looking for security. They are susceptible to overestimating the risk of investments.
- Active investors are willing to trade some of that security in exchange for higher gains. On the other hand, they are more prone to a behavior that displays overconfidence, which can lead them to underestimate risk. However, whenever they detect a loss of control, their risk tolerance declines rapidly.

Figure 1: Barnewall's Model



1.2 Bailard, Biehl, and Kaiser Model (BB&K)

This behavioral model is a five-way model, introduced in 1986, and is more complex than Barnewall's model. In this model, investor personalities are classified along two axes: **level of confidence and method of action**.

- **Level of confidence** is meant to categorize how much confidence an individual displays when making decisions. An individual with low confidence is labeled as anxious. At the opposite of the spectrum, individuals with high confidence level are defined as confident. There are no intermediate levels, but a financial advisor could notice that a specific individual may fall in between the two extremes.
- **Method of action** is intended to be used to identify whether a specific individual is methodical, analytical, or careful in the way they make decisions or whether they tend to be more on the spontaneous side. An individual who is very methodical and analytical is considered to be careful.

At the opposite end of the spectrum, an individual who is mostly spontaneous is defined as impetuous.

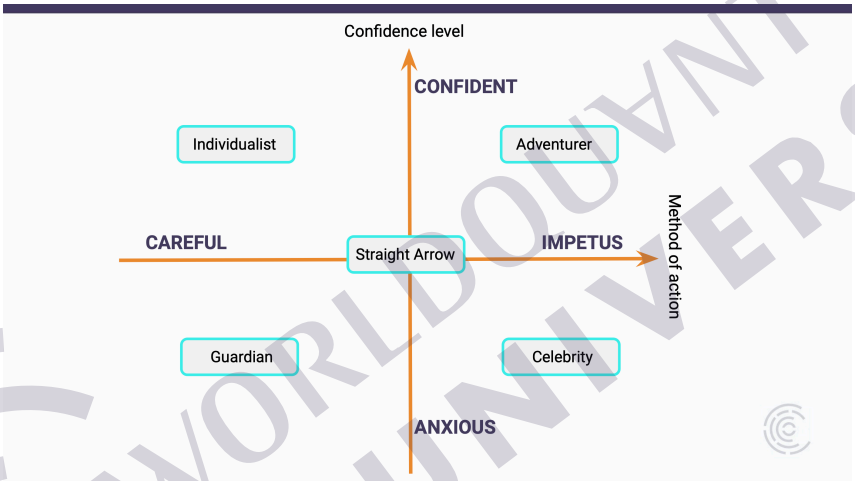
The BB&K model, based on these two dimensions, identifies five different personalities of investors (hence the five-way model):

- Straight arrow;
- Celebrity;
- Adventurer;
- Individualist;
- Guardian.

The fancy names are just that; they are supposed to help associate an image with a client's behavioral type.

At this point, a plot and a synopsis of the model (due to Kaiser) are very useful:

Figure 2: BB&K Model and Its 5 dimensions.



Straight arrow investors can be defined as sensible and secure as they fall in the center of the graph. They are the most balanced of the investor types and are willing to trade some risk in exchange for a suitable amount of additional return.

Guardian investors are careful and anxious. It is a type that is most often associated with individuals who are approaching retirement. Their number one concern is to protect their assets since they have less time to recover from bad investments.

Individualist investors are confident and tend to act in an independent manner. They prefer to make their own decisions after careful analysis and tend to process information rationally.

Celebrity investors have low confidence: They are anxious in fact and tend to be impetuous in the way they act. In practice, this means that we are in the presence of individuals who are aware of their limitations but who act in a spontaneous manner and enjoy being the center of attention. They are not known for being patient.

Adventurer investors are confident and impetuous. They are very difficult to deal with because they are very confident and do not shy away from taking chances. It is not unusual that adventurer

investors are usually persons with an entrepreneurial past and who were successful in their business.

What does all this mean for a financial advisor?

Straight arrow investors, guardian investors, and individualist investors are amenable to advice because of their careful behavioral component. However they take advice differently. The guardian type is very willing to listen to advice, especially if they recognize that the advisor has more experience. Individualist investors tend to act on their own when it is time to make decisions.

Celebrity investors are usually seeking out advice, but they may have strong opinions that need to be educated and possibly moderated or corrected.

Adventurer investors are the most difficult of them all to deal with. Highly confident and more than willing to take chances, they are reluctant to take advice.

1.3 Pompian Model

This model was developed in 2008 and it is a four-way model in the sense that it has four behavioral investor types (BITs) at its core: the **passive preserver**, the **friendly follower**, the **independent individualist**, and the **active accumulator**.

The four BITs are identified as such after an interview with clients and can be described as follows:

A **passive preserver** is an individual who places most emphasis on financial security and displays a conservative approach to investing. This investor is generally satisfied with the status quo and believes that preserving one's wealth is preferable to potentially higher returns if those come at a higher risk of losing money. In general, these investors are not prone to either frequent portfolio rebalancing or to excessive trading in and out of positions. Their actions tend to be driven more by how they feel and think, hence they are characterized more by emotional biases rather than cognitive ones. Preservers display mental accounting and accumulate wealth by investing carefully and in a balanced manner.

As preservers have high aversion to risk, they can panic easily and sell out of positions after suffering losses, which often causes them to miss the rebound. Furthermore, their low-risk and cautious approach to investments can produce portfolios that are heavily invested in bonds and cash instruments. Unfortunately, this could be detrimental to them achieving their financial goals.

A **friendly follower** is an individual with low to moderate risk tolerance and who does not have a keen interest in the financial markets. Therefore, these individuals tend to follow the advice of experts or peers when it comes to making investment choices and their decisions are driven by cognitive biases rather than emotional ones. One of their characteristic traits is that they are attracted by the latest investment trends for fear of missing out on something good and, as a consequence, exhibit a faulty understanding of the correct risk they are assuming and the magnitude of potential losses.

They are not likely to indulge in excessive trading in and out of positions or change their portfolio compositions often. The major danger they face is a consequence of their tendency to chase the newest investment advice or trends when they see or hear their peers doing it. As they are followers, they are likely to get into these investments in later stages as they fear missing out on the opportunities and do not really pay attention to whether the timing is right or not. In general, these

individuals can be educated on the benefits of portfolio diversification and the advantages of sticking to a long-term financial plan.

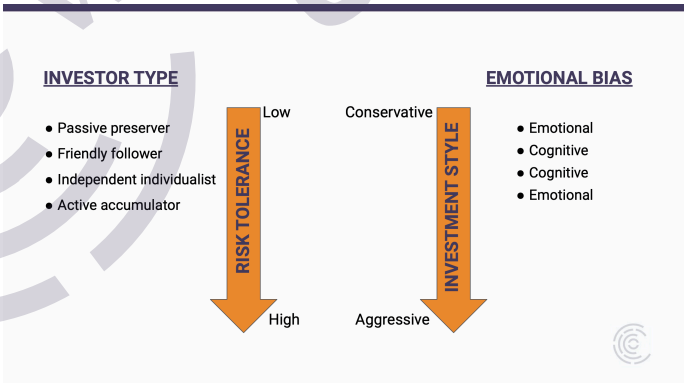
An **independent individualist** is a suitable description for a strong-willed, analytical, and critical thinker who also entertains unconventional ideas about investing. This group of investors is actively engaged in the investment process: They are usually not afraid to do their own research and understand the risks of their investments. They are willing to tolerate risk since they are confident about their ability to research the pros and cons effectively and are not afraid to put in the hours to research their investments.

Overall, we could refer to them as thinkers and doers as opposed to dreamers and followers. However, as they tend to act independently, they are susceptible to confirmation bias in the sense that they tend to ignore certain angles of the problem that run contrary to their views. They also are prone to falling in love with their ideas and are not open to entertaining new ideas and views. Because of this, they would benefit from discussions with their financial advisors as a form of check and possibly a final revision of some of their investment plans.

An **active accumulator** is a description reserved for individuals who make wealth accumulation a priority and are highly engaged in the process. Overall, these individuals possess a strong will and are both decisive and confident in making and following their investment choices, respectively.

These individuals can be recognized for having an entrepreneurial mindset. Often, they have already been successful in their field and are aware of the research and hard work necessary to succeed and apply that approach to investing as well. Nevertheless, their high confidence lvel also causes them to be susceptible to emotional biases like overconfidence and self-control as they dive into excessive trading and portfolio rotations. They seem to enjoy the thrills associated with making good investments and, as a consequence, are attracted to high-risk investments and do not pay enough attention to either asset allocation or diversification.

Figure 3: Pompian Model and Its 4 BITs



Regarding the main types of behavioral biases within each BIT, the most relevant are:

- **passive-preserver**: endowment, loss aversion, status quo, regret aversion, mental accounting, anchoring.
- **friendly follower**: regret aversion, availability, hindsight, framing.
- **independent individualist**: overconfidence, self-attribution, conservatism, confirmation, representativeness, availability.
- **active accumulator**: overconfidence, self-control, illusion of control

Are behavioral types the last word for BPT? The answer is that while they do help, they are not without limitations. In fact, it is possible that one investor displays the traits of more than one type, which makes it difficult to assign them to a specific group. Furthermore, it is not uncommon for individuals to display both emotional and cognitive biases at the same time. And, if that were not enough, the age of investors also plays a role. It is quite admissible that, as investors age, whether they realize it or not, their BIT changes. A portfolio manager or financial advisor should also be careful not to use the models above as if they were strict mathematical models: Two individuals could be classified in the same group, but they may still need different types of advising.

2. Momentum, Fat Tails, Bubbles, and Crashes: The Behaviorists' Explanations

Traditional finance believes that arbitrage has the power to quickly erase any unjustified mispricing —so then why do we have bubbles?

BF suggests that individual investors display an eagerness to imitate the decision of others, a situation called **herding**. However, just because many do it, the fact that many investors exhibit the same behavior does not imply that said behavior is rational.

This behavior can lead to *bubbles* and *crashes* in the most extreme cases. In milder cases, it could explain why stock returns have fat tails. This behavior has been observed more often in the case of illiquid assets where trading is less frequent and investors believe that changes in prices may be a proxy for someone having access to non-public information instead of being caused by lack of liquidity.

Momentum arises when price movements are correlated with their past history. Several studies have highlighted the fact that over shorter periods (1–2 years), there is positive correlation, which, however, tends to disappear and even become negative over longer periods (2–5 years) as prices revert to their means.

Momentum has been invoked to partly explain short-term underreaction to relevant information and longer term overreaction.

The behavioral bias that causes it is anchoring. In fact, investors attach a special significance to the price at which they bought the asset. When they believe in mean reversion, they are initially underreacting to new information and choose to believe that the new higher price has actually introduced new higher risk as well. Hence, it takes time for stock prices to adjust to new positive information.

According to behaviorists, the anchoring bias is further reinforced by availability bias, which in this particular instance could be labeled as a **recency effect**. When investors see that a stock goes up for several days in a row or even over a period of weeks, they extrapolate that the stock will continue to go up in the future as well and thus do not feel a particular sense of urgency to buy as they are essentially seeing a trend that may or may not be there.

After a period of time where investors have been on the sidelines, they start to suffer from regret aversion, the feeling that they might be missing an opportunity or that everyone else is taking advantage of a great opportunity and they are among the few who have not done so yet. During this period, there is also some hindsight bias as investors who have not bought the asset blame

themselves for not having noticed the development and having failed to realize what was a very easy-to-spot trend. At this point, feeling regret and blaming themselves for missing the trend, they rush to remedy the issue. This action creates a **trend-chasing effect** where investors rush to buy assets they wish they had owned already.

At some point, evidence collected over the years has shown that over longer periods of a few years (3–5), mean reversions take place.

When corrections begin, investors experience loss aversion and therefore hold on to stocks that are losing value. This behavior is also called a **disposition effect**. Psychologically, they hope that the loss in value is just a temporary effect and that soon assets will start to go up again, and this attitude slows down the correction process. We can recognize some anchoring bias as well while investors try to rationalize the losses. Players like hedge funds could even be trying to sell short to capitalize on what is perceived as a temporary correction at best.

Eventually, particularly in the case of bubbles, as the bubble unwinds, investors begin to realize what is going on and capitulate. Now, fearing to be among the last ones still holding a bad asset, they rush to sell and accelerate the price decline. In particular, since we mentioned hedge funds, when the average investors see hedge funds selling short assets, they are susceptible to think that hedge funds managers must know something they don't and that points to prices falling further. Thus, this particular sort of negative momentum can also trigger a rout.

Note: TF has tried to explain momentum without using behavioral biases. Unfortunately, the bibliography is quite extensive, and this is not the place to get into it. We mention that there is also an extension of the Fama French three-factor model, called the **Fama French and Carhart model,** in which momentum is added as an additional factor to the model.

3. A Working Example

Thunder & Bolts & Associates (TBA) is an investment management firm that has been in operation since 1985 and uses proprietary valuation models based on fundamental analysis to select individual stocks and bonds for the actively managed portfolios of their clients. During the last 10 years, TBA has been taking into account investors' behaviors and past experiences when advising its clients.

Muinul Chowdhury is a young professional who recently earned a M.Sc. degree in Finance from WQU and has since been hired by TBA in their portfolio management team. He has been assigned three new clients whose portfolios need to be transitioned to TBA's managed accounts. To this purpose, Mr. Chowdhury is reviewing the completed questionnaire for each of them and prepares three short bios:

Colby Skelton, 40 years old, earns his income from royalties from his successful travel books. He has accumulated a portfolio with a current value of $3.0 million. Up to now, Mr. Skelton has managed his own portfolio and feels very confident about his own ability to invest successfully when good opportunities present themselves. Still, in several instances, Mr. Skelton's portfolio contained positions with sizable losses that he proved reluctant to sell. As a matter of fact, Mr. Skelton stated that, due to the size of the losses and the size of his portfolio, he decided to seek professional help. He acknowledges that he would be agreeable to considering high-risk investments as long as his research indicates that these investments represent an attractive

opportunity. Currently, his portfolio consists of 12 equity positions of different dollar values, and these equities are diversified across seven different industries and 4 different countries.

Jayne Griffith is 60 years old and has steadily risen through the ranks of a government agency after working there for nearly 30 years, and she is a now a deputy director. She currently has a financial portfolio valued at $1.2 million. Mrs. Griffith has little experience in investing and has come to TBA as she is nearing retirement, and about 75% of her portfolio is invested in a single company that her uncle started and manages. She is specifically worried that an economic downturn could affect the company and her retirement plans. She stated also that, at this stage of her life, she would prefer to avoid any high-risk investments.

Neal Johnson is 50 years old and has been very successful starting his own high tech company when he was only 22. He found a very small niche for his company with little competition that he has exploited very well. As a result, Mr. Johnson earns several million dollars each year. So far, Mr. Johnson has managed his own portfolio via an online brokerage account. He invested about $20 million in it and his portfolio currently has a value of $19 million. He admits to having limited investment expertise and, despite his high-tech business, he considers himself a limited to moderate risk taker when it comes to investing his money. He explains that he has been selecting his investments mainly by acting on the advice of other managers in his field, from what he learned by attending sector conferences, and by taking advantage of opportunities he hears about at the many social events he attends. Mr. Johnson has purchased almost exclusively the most popular stocks for his portfolio and is invested completely in shares of 10 high-tech U.S. companies. Mr. Johnson complains that, due to his very busy schedule, he has often missed the right time to buy and sell stocks or, otherwise, he would have "made a killing" in the stock market.

Question: Which of the 3 clients is most likely a friendly follower?

Neal Johnson. This can be determined from the following elements:

- He stated that he has a low to moderate risk tolerance;
- he follows the advice of other high-tech professionals, friends, and acquaintances and possibly financial advisors; and
- the majority of his investments are in popular assets without checking whether these are truly suitable for his long-term goals.

Home bias is defined as a bias where an investor prefers investing in their domestic country vs. foreign countries or prefers to invest in companies with which they are familiar over companies that they do not know. **Familiarity investment** (bias), instead, is defined as the preference to stay in one's comfort zone, i.e., for instance investing in a sector in which one has invested before. **Naive diversification** is when an investor allocates approximately the same amount to each asset class. Finally, another behavioral bias that is often mentioned in the field is **gambler's fallacy,** which can be described as a situation in which an individual believes that a random event is less likely to happen following an event or series of events.

Question: Considering the following biases: home bias, loss aversion, naive diversification, and familiarity investment, what is the most likely behavioral bias for Mr. Skelton, judging from his portfolio? And what would you say the least likely bias is?

The most likely bias is loss aversion. The least likely is home bias. The reasoning is that:

- Mr. Skelton has displayed reluctance to sell positions that have lost value, which is evidence of loss aversion.
- Mr. Skelton's portfolio contains equities of four different countries, hence he is not displaying home bias.
- Mr. Skelton is willing to consider new investments, even high risk investments so there does not seem to be a case for familiarity investment. But this is less clear from the information we have available.
- Mr. Skelton does not display naive diversification as his allocations have different dollar values. As such, it is difficult to agree on his attitude towards naive diversification.

Patrick Hess, Ph.D., CFA is a senior portfolio manager at TBA and has had a long successful performance record. He has voiced his concern about using BIT models as they have many limitations. He is also supervising Mr. Chowdhury and explains to him that, although TBA uses fundamental research and it provides great value to its customers, technical analysis is also compatible with sound investment advice. Upon hearing this statement, another colleague, Devin Wirawan, CFA, feels compelled to jump into the discussion and states that TBA fundamental value models incorporate hundreds of pieces of information, and as such, they are more sophisticated and thorough than any other analysis. Finally, he says that he believes that the opportunities identified with technical analysis are actually not real market anomalies, but rather the result of under-estimating true risk exposure. Dr. Hess counters by saying that technical analysis has the potential to detect opportunities in cases where markets are over or underreacting to new relevant information. Mr. Chowdhury then states that from what he has seen in his still relatively short career, every time a sector falls below its 10-year average level, the sector has recovered to its mean in a relatively short period of time.

Question: Concerning Dr. Hess' statement regarding behavioral types, it is most likely justified because individual investors

1. exhibit characteristics of several investor types;
2. retain the same behavioral biases as they age;
3. display primarily emotional or cognitive biases, but not both.

The answer is 1; 2 and 3 are clearly wrong.

Question: Based on Mr. Wirawan's tendency to strictly adhere to TBA's proprietary valuation models, he is most likely to exhibit overconfidence in TBA's models because of:

1. availability bias
2. self-attribution bias
3. an illusion of control

The most likely explanation is 3. Mr. Wirawan's belief that, through the model he uses, he is in a position to use new information better than his competitors. The fact that the models use a lot of data could actually reinforce his illusion of control. While, in general, more information is better than less information, the use of complex models does not necessarily say a lot about how accurate the forecasts have been in the past. There is really no indication of availability bias and Mr. Wirawan is not self-attributing the success of TBA's models.

Question: Regarding technical analysis, which of the three individuals has made the most accurate statement?

Given the three statements, Mr. Chowdhury made the most accurate statement. What he described is momentum investment and there is a certain amount of evidence that markets display short-term underreaction and long-term overreaction to information.

4. Seven Sins of Fund Management

Individual investors, analysts, and portfolio managers have one thing in common: They are all first and foremost individuals and, as such, they display behavioral biases. When we invest, we do not need to prove our ideas in a mathematical sense, and therefore, we tend to indulge in faulty reasoning mixed up with a lot of feelings.

While BF has been around over 40 years, it was not widely known among practitioners outside of academia. That began to change in the late 1990s and even more so in the early 2000s.

In 2005, James Montier an economist at Dresdner Kleinwort Wasserstein in London—later co-Head of Global Strategy at Société Générale, and, more recently, member of GMO's asset allocation team, Montier has often been voted a top-rated strategist in the annual Thomson Extel survey—published a research note, **Seven Sins of Fund Management: A Behavioral Critique**, which flew off the shelves: It had to be reprinted many times over a period of several months.

In a somewhat iconoclastic style, Mr. Montier provided a very effective way to analyze and think of the most egregious behavioral traits that often impede most individuals (day traders, portfolio managers, financial analysts, etc.) from being truly good at investing. The table that follows is an abridged version of Montier's Research Report from 2005.

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"SIN"	DESCRIPTION
Forecasting (Pride)	There is a large body of evidence that people are simply not good at forecasting. However, about 75% of fund managers think that they are above average at their jobs! The inability to forecast seems to be rooted in the fact that we all seem to be over-optimistic and over-confident.
Illusion of Knowledge (Gluttony)	Like the robot Johnny 5 in the 1986 movie *Short Circuit*, individuals thirst for more and more inputs/information. Investors display the belief that they need to know more than everyone else in order to outperform. This is probably a consequence of having been taught from an EMH point of view. Unfortunately, the verdict from the field seems to be that increasing the amount of information gathered serves mainly to increase our confidence rather than increase our accuracy.
Meeting Companies (Lust)	Analysts meet with companies' management. Individual investors read their research reports or companies' reports. There are at least five hurdles that must be overcome if meeting companies is to add value to an investment process: 1. More information is not necessarily better information. 2. Corporate managers are not perfect and have their own behavioral issues, often in the form of cognitive illusions. 3. We suffer from confirmation bias and tend to look mostly at the information that agrees with us and discard the rest. 4. We naturally tend to obey figures of authority. Top management, being at the top, inspires such feelings. Hence, analysts and fund managers are effectively over-awed and take their words for it. 5. We are lousy at discerning truth from deception, although we tend to believe that we can spot liars.
Thinking you can outsmart everyone (Envy)	Over-optimism and overconfidence are likely the main culprit if we are prone to thinking along the lines of "Now I understand Behavioral Finance; I can outsmart everyone else." Plenty of experiments have shown that we are much worse than we think.
Short time horizon and over-trading (Avarice)	Too many investors confuse noise with news and try to outsmart each other and end up with time horizons that are too short (usually 11 months for stocks on NYSE at the time of the research). Over such a short time span, returns have nothing to do with intrinsic value or discounted cash flows and are just a function of price changes.
Believing everything you read (Sloth)	Stock brokers spin stories, which act like sirens drawing investors onto the rocks. We all love a story and we appear to be hardwired to accept stories at face value, even some very improbable ones.
Group based decisions (Wrath)	We tend to accept a principle according to which groups are better at making decisions than individuals. Ideally, groups meet to exchange ideas and reach sensible conclusions. Yet, research by psychologists over three decades has shown that groups' decisions are usually among the worst decisions. In fact, groups, instead of offsetting each others' biases, end up amplifying the problem! Groups tend to reduce the variance of opinions and for that reason members have more confidence in their decisions (**social proof**). Members enjoy enhanced competency and credibility in the eyes of their peers and do not like to provide information that runs contrary to the group's consensus. Think of analysts for whom producing forecasts too far away from their peers is very dangerous as they could lose their jobs if they are off. Hence, they tend to stay close to peers' forecasts very often. These forecasts are then passed on to investors or portfolio managers.

So, the lesson is that everyone involved in investing money—analysts, portfolio managers, and individual investors alike—suffers from behavioral biases that compromise one's ability to use information properly and to avoid the many traps that, to use James Montier's words, attract us onto the rocks.

5. Conclusion

This lesson was mainly about behavioral models and how they help financial advisors and portfolio managers to know their clients and to build a better relationship with them.

The next module will consider the problem of finding a better portfolio construction technique than mean-variance optimization and will look at the drawbacks of the optimal Sharpe portfolio. In addition, it will investigate what investment strategy is best aligned with a logarithmic utility function and what investment strategy results in maximum wealth.

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